

PHYSICAL RISK · SOUTH ASIA · AGRICULTURAL FINANCE

The Financial Anatomy of a Monsoon Deficit: What Happens to Global Markets When India's Rain Fails

A new El Niño is forming in the Pacific. NOAA data shows Niño 3.4 sea surface temperatures rising ahead of the pace of recent warming years. Across 150 years of India Meteorological Department records, the same physical sequence follows every time — and its financial consequences reach commodity markets, agricultural credit portfolios, sovereign balance sheets, and corporate earnings across six continents.

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August
2026

12 min
read

Sources: IMD · FAO · IPCC AR6 ·
RBI · NGFS Phase 4

The India Meteorological Department has measured monsoon rainfall since 1871. Across 150 years of instrumental records, the correlation between El Niño events and below-normal monsoon rainfall holds with a consistency that climate scientists describe as one of the most reliable teleconnections in the global climate system.

[IMD — 1871–2024 series](#)

A new El Niño is forming now. Sea surface temperatures in the Niño 3.4 region — the central equatorial Pacific index that defines El Niño events — are rising ahead of the pace recorded in other recent warming years. [NOAA ENSO Monitoring](#) The trajectory points toward a significant event by late 2026 and into 2027.

This piece maps what that means for India's monsoon, and what India's monsoon means for commodity markets, agricultural credit, sovereign fiscal stress, corporate earnings, hydropower, and emerging market debt.

What the Indian Summer Monsoon Actually Is

The southwest monsoon arrives over Kerala in early June, advances northward through July, and withdraws by late September. Four months. That window delivers the rainfall that determines whether India's agricultural year succeeds or fails. The IMD's Long Period Average — computed over the 1971 to 2020 reference period — is 87 centimetres for the full season. [IMD LPA 1971–2020](#)

SOURCE: INDIA METEOROLOGICAL DEPARTMENT — LONG PERIOD AVERAGE SERIES (1971–2020)

*"The southwest monsoon season (June to September) accounts for approximately **70 to 75 percent of the country's total annual rainfall**. The Long Period Average (LPA) of All-India rainfall for this season is 87 centimetres, computed over the 1971–2020 reference period. Departures from the LPA of more than 10 percent below normal are classified as deficit years. Departures exceeding 19 percent are classified as large deficit — a threshold breached in 2002 and approached in 2009."*

70%

of India's annual rainfall
delivered by the monsoon in
4 months

India Meteorological
Department — LPA Series

40%

of global rice trade supplied
by India

USDA Foreign Agricultural
Service

800M

people covered by India's
National Food Security Act

Ministry of Consumer
Affairs, India

Agriculture accounts for approximately 15 percent of India's GDP [MoSPI India](#) — but it employs close to 50 percent of the workforce and underpins the rural

purchasing power that drives demand across consumer staples, tractors, motorcycles, and financial services. When the monsoon underdelivers, the economic transmission is broad and deep.

The Physical Mechanism: How El Niño Breaks the Monsoon

The Indian summer monsoon is driven by the temperature differential between the rapidly warming Asian landmass and the cooler Indian Ocean. As land heats faster than sea, it draws moisture-laden ocean air inland — creating the sustained low-pressure system that generates rainfall across the subcontinent.

El Niño disrupts this by weakening the Walker Circulation — the large-scale atmospheric loop that normally carries warm, moist air westward across the tropical Pacific, drives convection over the Indian Ocean, and reinforces the pressure gradient that pulls the monsoon inland. **IPCC AR6 WGI Ch.3** When Pacific SST anomalies rise, convection migrates eastward, the Indian Ocean loses its moisture driver, and the monsoon weakens.

SOURCE: IPCC AR6 WORKING GROUP I – CHAPTER 3 (ENSO AND GLOBAL TELECONNECTIONS)

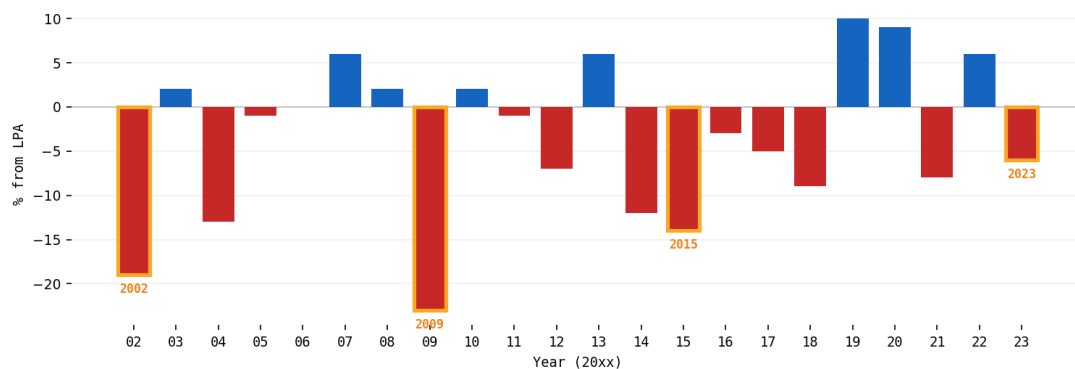
"The suppression of the Indian summer monsoon during El Niño events is among the most robust and historically consistent teleconnections in the global climate system.

ENSO-driven weakening of Walker Circulation reduces moisture transport into the Indian subcontinent, producing below-normal all-India rainfall in moderate to strong El Niño years with high historical reliability across the instrumental record. This teleconnection is projected to persist under all warming scenarios assessed."

The chart below shows all-India monsoon rainfall departure from the long-period average from 2002 to 2023, using IMD data. El Niño years are highlighted. The pattern is not ambiguous.

INDIA ALL-INDIA MONSOON RAINFALL — DEPARTURE FROM LONG-PERIOD AVERAGE

% deviation from LPA · Source: India Meteorological Department · Amber bars = El Niño years (2002, 2009, 2015, 2023)



SOURCE: IMD SEASONAL MONSOON REPORTS — KEY EL NIÑO YEARS

2002: All-India rainfall 19% below LPA. Moderate El Niño. Drought declared in 14 states. Agricultural GDP growth fell to near zero. **2009: 23% below LPA — most severe deficit recorded in 27 years.** Moderate El Niño with Niño 3.4 anomaly approximately +1.0°C. Drought declared across 10 states and 278 districts. **2015:** 14% below LPA. First year of 2015–16 Super El Niño. Kharif output compressed across Maharashtra, Karnataka, and Uttar Pradesh.

The 2023 Case: From SST Anomaly to Rice Price Spike

The 2023 to 2024 Super El Niño peaked at approximately +2.0°C above the 1991 to 2020 baseline in the Niño 3.4 region. [NOAA ENSO Monitoring](#) India's 2023 southwest monsoon was spatially uneven and below normal in several major kharif-producing states. [IMD Seasonal Report 2023](#) In July 2023, India imposed export restrictions on non-basmati white rice. In August 2023 the restrictions widened.

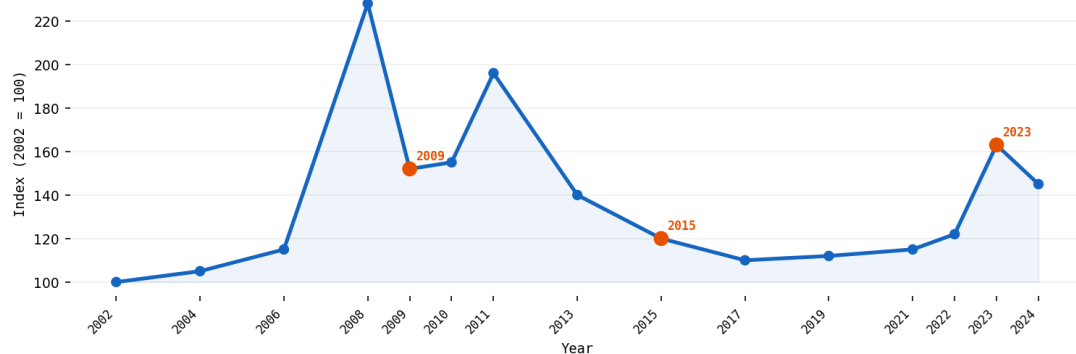
[Ministry of Commerce, India](#)

SOURCE: FAO RICE MARKET MONITOR — SEPTEMBER 2023

"Following India's announcement of export restrictions on non-basmati white rice in July 2023, **international rice prices surged to their highest levels since 2008** — a 15-year high. The FAO All Rice Price Index rose by more than 25 percent within two months of the restriction. Countries sourcing the majority of their rice imports from India — across Sub-Saharan Africa, Southeast Asia, and the Middle East — faced immediate supply substitution challenges. India supplies approximately **40 percent of global rice trade**. No other single exporter can substitute at scale."

GLOBAL RICE PRICE INDEX — 2002 TO 2024

2002 = 100 · Source: FAO Rice Market Monitor · Orange dots = El Niño years · Key events annotated



"India supplies 40 percent of global rice trade. When India partially exits the export market, the price signal reaches every food-importing country within weeks — from Bangladesh to Senegal."

Six Financial Transmission Pathways

The physical parameters do not stay in meteorology. A monsoon deficit moves through six channels into financial statements, credit portfolios, and sovereign

balance sheets. The intensity bars below show relative financial impact in a deficit El Niño year, with the primary data source for each pathway.

FINANCIAL TRANSMISSION PATHWAYS – RELATIVE IMPACT, EL NIÑO MONSOON DEFICIT YEAR

Sources: RBI FSR · FAO · CEA India · IMF GFSR · NABARD Annual Reports · Company filings

Agricultural Commodity Markets

FAO Rice Market Monitor

Rice, sugar, cotton – direct supply disruption when monsoon fails in key growing states. Simultaneous across multiple crops in a single season.



Agricultural Credit Risk

RBI Financial Stability Report

Gross NPA ratios in agri segment of scheduled commercial banks rise 50–100 bps above trend in deficit years. Stress concentrated in rain-fed districts.



Sovereign Fiscal Stress

Ministry of Consumer Affairs, India

Food Corporation of India draws buffer stocks or procures at higher MSP. 800M NFSA beneficiaries mean subsidy costs rise when production falls.



FMCG Corporate Earnings

Company earnings filings (HUL, ITC, Dabur, Marico)

Rural India is 35–40% of FMCG revenues. Monsoon is the single largest variable cited in quarterly results – earnings impact visible within 2 quarters.



Hydropower and Energy Substitution

Central Electricity Authority of India

25% of India's electricity from hydro. Monsoon deficit reduces reservoir levels, curtails generation, forces thermal coal substitution and increases import bill.



Emerging Market Contagion

IMF Global Financial Stability Report

Food-importing sovereigns across Asia, Africa, and the Middle East face import cost spikes. IMF GFSR documents monsoon-linked EM sovereign stress in major deficit years.

SOURCE: RESERVE BANK OF INDIA – FINANCIAL STABILITY REPORT, JUNE 2023

"Gross non-performing assets of scheduled commercial banks in the agriculture and allied activities segment **tend to worsen in years of below-normal monsoon**, as reduced agricultural income affects the repayment capacity of farm borrowers. Monitoring data shows NPA stress concentrated in rain-fed districts experiencing rainfall deficits of more than 10 percent below the long-period average. In such districts, agricultural loan recovery falls materially in the harvest quarter following the deficit season."

SOURCE: CENTRAL ELECTRICITY AUTHORITY OF INDIA – ANNUAL REPORT

"Hydropower contributes approximately **25 percent of India's total installed generation capacity**. Reservoir-based generation is directly dependent on monsoon rainfall for recharge. Below-normal monsoon years reduce average reservoir levels at key projects, curtailing generation during the post-monsoon period and increasing dependence on thermal generation, which in turn increases coal procurement requirements and generation costs across the grid."

What Climate Science Says About the Monsoon Through 2050

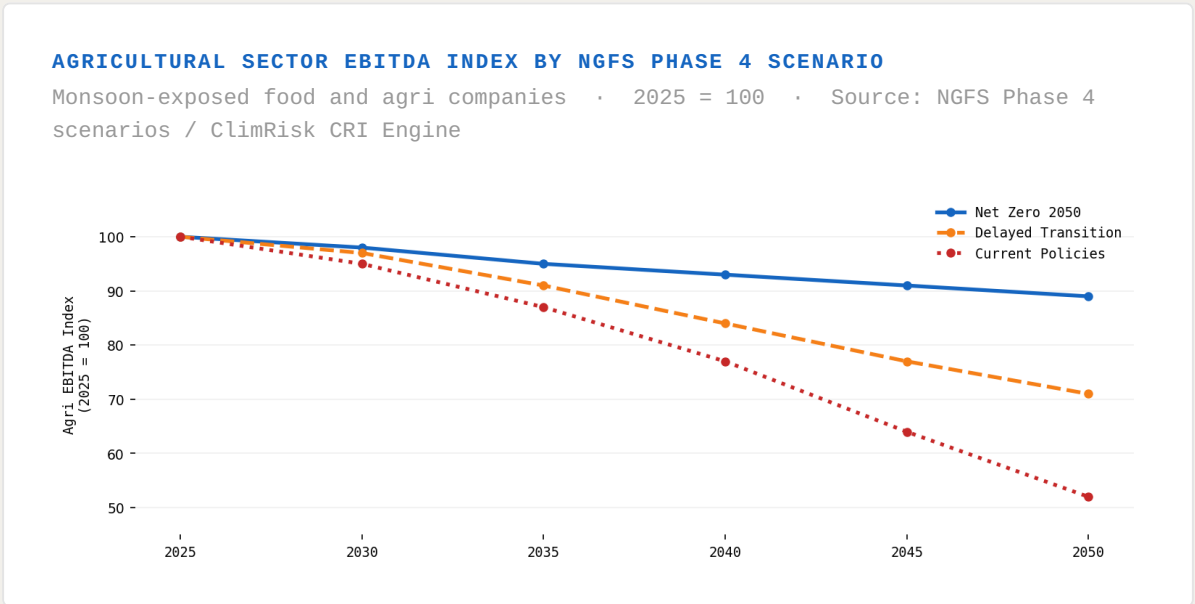
SOURCE: IPCC AR6 WORKING GROUP I – CHAPTER 8 (WATER CYCLE CHANGES UNDER CLIMATE CHANGE)

"Global monsoon precipitation is projected to intensify with warming, but with **higher interannual variability and greater extremes**. Wet years are projected to become wetter and dry years drier across most monsoon regions. The amplitude of these swings increases with baseline warming — meaning that El Niño-driven monsoon deficit years produce progressively larger agricultural disruptions as global mean temperature rises, even with no change in El Niño frequency."

El Niño events are not projected to become more frequent. [IPCC AR6 WGI Ch.3](#) The ENSO cycle does not systematically accelerate under climate change. What changes is the baseline on which each event operates. A Niño 3.4 anomaly of +1.0°C in 2035 operates on an ocean that is already 0.3 to 0.4°C warmer than the same anomaly encountered in 2009. The amplification of agricultural disruption comes not from more frequent events but from a warmer baseline beneath each event.

The NGFS Phase 4 scenarios translate this science into financial projections.

[NGFS Phase 4 Documentation](#) The chart below shows the modelled agricultural EBITDA index for monsoon-exposed food and agri companies under three scenarios through 2050.



El Niño Deficit Year – Impact	Net Zero 2050	Delayed Transition	Current Policies
Rice and commodity supply	MODERATE	HIGH	SEVERE
Agricultural NPA pressure	MODERATE	HIGH	SEVERE
Sovereign food subsidy cost	MODERATE	HIGH	SEVERE
FMCG rural revenue impact	MODERATE	HIGH	HIGH
EM sovereign credit stress	MODERATE	HIGH	SEVERE

*"Under Current Policies, **global mean temperature rises approximately 2.5 to 3°C above pre-industrial by mid-century.** Under this pathway, El Niño events from the 2030s onward operate on a structurally warmer baseline, amplifying the physical disruption to the South Asian monsoon and associated precipitation patterns. Under Net Zero 2050, warming stabilises and the amplification effect is contained near current levels."*

What This Means for Capital Allocation

Standard financial models do not price monsoon risk systematically. India equity research treats the monsoon as a one-line qualitative comment in FMCG and agri sector notes. Agricultural credit risk frameworks do not embed ENSO cycle probability distributions into NPA forecasting. Sovereign credit models do not connect monsoon deficit probability to fiscal stress estimates under different warming pathways.

The 2023 case makes this gap concrete. The physical cascade from Niño 3.4 SST anomaly to India rice export restriction to the FAO-recorded 15-year price high

[FAO Rice Monitor, Sept 2023](#)

was physically predictable months before it occurred.

The Walker Circulation signal was visible in atmospheric data by April 2023.

[IPCC AR6 WGI Ch.3](#)

IMD's June 2023 forecast flagged uneven spatial distribution.

[IMD Seasonal Forecast, June 2023](#)

The financial consequences arrived in July. Markets priced them as a surprise.

"The physical signal was visible in April. The markets reacted in July. That gap — between

observable atmospheric physics and financial pricing — is where the loss occurs."

The physical parameters are public. The atmospheric physics is understood. The IMD publishes 150 years of rainfall data. [IMD](#) The FAO publishes monthly commodity price monitoring. [FAO](#) The RBI publishes semi-annual financial stability assessments. [RBI FSR](#) The IPCC publishes the science. [IPCC AR6](#) What is missing is the analytical framework that connects the physical signal to the financial output — at the asset level, before the cascade completes rather than after.

CLIMATE RISK INTELLIGENCE

ClimRisk — CRI Engine

The CRI engine maps asset and supply chain footprints against physical hazard projections — WRI Aqueduct 4.0, NASA NEX-GDDP CMIP6, and NOAA ENSO cycle data — and translates monsoon risk into financial terms: EBITDA compression, enterprise value impact, credit loss probability, across three NGFS scenarios through 2050. Not a risk score. A DCF with physical climate adjustment terms. 48-hour turnaround at asset level.

climrisk.io

Data sources: India Meteorological Department Long Period Average series and seasonal monsoon reports · NOAA ENSO Monitoring Centre · FAO Rice Market Monitor (September 2023) · FAO Sugar and Sweeteners Market Assessment · USDA Foreign Agricultural Service production and trade estimates · Ministry of Statistics and Programme Implementation, India · Reserve Bank of India Financial Stability Reports (June 2022, June 2023) ·

NABARD Annual Reports · Central Electricity Authority of India Annual Report · IPCC AR6 Working Group I Chapters 3 and 8 · NGFS Phase 4 scenarios (Network for Greening the Financial System) · IMF World Economic Outlook and Global Financial Stability Report

Shrinivash D Kannan is Founder of ClimRisk, a climate financial risk intelligence firm. The CRI engine translates physical climate hazard and NGFS transition scenarios into DCF financial outputs – EBITDA, enterprise value, credit loss probability – across a 25-year horizon.